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SUPPLEMENTARY MATERIAL

The BBP Formula and the Coupling Polynomial on One Ladder

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Abstract

The coupling polynomial $f(K,s) = 1 + sK + K^2$ of Gravity Field Dynamics takes the values $\Phi_6(K), \Phi_4(K), \Phi_3(K)$ at $s = -1, 0, +1$, so its three states form one row of the ladder $X \to X^2$ of cyclotomic polynomials. We show that the Bailey–Borwein–Plouffe (BBP) formula for π lies on the same ladder, in its Field Origin column. In the variable $u = y/\sqrt{2}$ the denominator of its integral is $16(1-u)(1+u)(1+u^2)(1+u^4)$, three rungs Φ_2, Φ_4, Φ_8, which are the Field Origin polynomial at three consecutive levels and take the values $5, 17, 257$ at $K = 4$. The rungs are the binary coordinates of the cube, the top one being the half-turn. Every formula for π of BBP's type must drop the rung at the coupling row's own level, Φ_4, together with one half of Φ_8, because their poles carry the primes 3 and 5; π is read from the other half of $\Phi_8 = \Phi_4(X^2)$, the Field Origin one level up. Silence on the TIME slot and closure of the weights both act on the one remaining factor, and both select BBP. Taking the algebra further: each of the seven constants of BBP's slots uses exactly the pole pairs whose arithmetic it contains, so the Field Origin pair is used only by $\ln 3$ and $\sqrt{2}\arctan(1/\sqrt{2})$; π lies on the single rung Φ_8, in the base -4 formula; at BBP's point $1/\sqrt{2}$ the rungs Φ_{2^k} take the values $F_{k-2}/2^{2^{k-2}}$, the Fermat numbers $3, 5, 17, 257, \ldots$, which are the Field Origin column read down the ladder, so that the rung Φ_{16} carries the Field Origin's own value 17; and climbing the ladder to periods 16, 32 and 64 adds no formula for π, because every new rung carries its own Fermat prime and must be dropped.

Keywords: BBP formula, cyclotomic polynomials, coupling polynomial, Field Origin, stellated octahedron, proof

Setting

Let D be the positive integer with $2(D+1) = 2^D$, so $D = 3$ and $K = D + 1 = 4$ [7]. The *coupling polynomial* is $f(K,s) = 1 + sK + K^2$, read at $s = +1$ (TetA), $s = 0$ (the Field Origin) and $s = -1$ (TetB). Φ_n is the n-th cyclotomic polynomial, the minimal polynomial of $e^{2\pi i/n}$.

The *slot series* are $S_j = \sum_{k\ge 0} 16^{-k}/(8k+j)$, $j = 1, \ldots, 8$. A *formula of type* $(*)$ for a number c is a rational vector w with $\sum_j w_j S_j = c$. Expanding $1/(16 - y^8)$ on $[0,1]$ gives

$$\sum_j w_j S_j = \int_0^1 \frac{p_w(y)}{16 - y^8}\, dy, \tag{1}$$

with $p_w(y) = 16\sum_{j=1}^{8} w_j\, y^{j-1}$. BBP is $w_{\mathrm{BBP}} = (4, 0, 0, -2, -1, -1, 0, 0)$ [1]. With $a = (8, -8, -4, -8, -2, -2, 1, 0)$, every formula of type $(*)$ for π is $w(r) = w_{\mathrm{BBP}} + r\,a$, $r \in \mathbb{Q}$ [2, 4]. The proof of that statement [4] writes each slot as

$$S_j = -\tfrac{1}{8} \sum_{\omega^8 = 1} (\sqrt{2}\,\omega)^j\, L_\omega, \tag{2}$$

with $L_\omega = \mathrm{Log}\left(1 - \bar{\omega}/\sqrt{2}\right)$, and shows, with Baker's theorem [3], that the seven logarithms $i\pi$, $\log 2$, $\log(1+\sqrt{2})$, $\log(2 \pm i)$, $\log(\sqrt{2} \pm i)$ in which every L_ω is expanded are linearly independent over the algebraic numbers. We use these two facts.

The Coupling Polynomial Is a Row of the Ladder

Theorem 1. *$f(X, +1) = \Phi_3(X)$, $f(X, 0) = \Phi_4(X)$ and $f(X, -1) = \Phi_6(X)$. Under $X \to X^2$,*

$$\begin{aligned} &\Phi_3(X^2) = \Phi_3(X)\,\Phi_6(X), \\ &\Phi_4(X^2) = \Phi_8(X), \qquad \Phi_6(X^2) = \Phi_{12}(X). \end{aligned} \tag{3}$$

So the rows (Φ_6, Φ_4, Φ_3) at $X = K^{2^n}$ are $(13, 17, 21)$, $(241, 257, 273)$, $\ldots$, and the Field Origin column $\Phi_4(K^{2^n}) = 1 + K^{2^{n+1}}$ is $17, 257, 65537$.

Proof. The roots of $X^2 \pm X + 1$ and $X^2 + 1$ are the primitive roots of unity of orders 3, 6 and 4. The square roots of a primitive n-th root are primitive $2n$-th roots when n is even, and one primitive n-th and one primitive $2n$-th root when n is odd. □

The cyclotomic form of f and this ladder are already in the archive [5]; Theorem 1 states them as the frame for what follows.

BBP Lies on the Field Origin Column

Lemma. *For base 2^{D+1} and period 2^D, the poles of $1/(2^{D+1} - y^{2^D})$ lie at radius $2^{(D+1)/2^D}$. The radius is $\sqrt{2}$, and the poles include $1 + i$, exactly when $2(D+1) = 2^D$.*

Proof. $(D+1)/2^D = 1/2$ is the stated equation, and $(1+i)^8 = 16$. □

Theorem 2. *With $y = \sqrt{2}\,u$,*

$$16 - y^8 = 16\,(1-u)(1+u)(1+u^2)(1+u^4), \quad (4)$$

and the rungs $1+u^{2^k}$, $k = 0, 1, 2$, are $\Phi_2(u), \Phi_4(u), \Phi_8(u)$, the Field Origin polynomial at three consecutive levels: $\Phi_2(X) = \Phi_4(\sqrt{X})$, $\Phi_4(X)$ and $\Phi_8(X) = \Phi_4(X^2)$. At $X = K$ they are $5, 17, 257$. Neither Φ_3 nor Φ_6 divides the denominator: of the coupling row, only Φ_4 occurs.

Proof. $1 - u^8 = (1-u)\prod_{k=0}^{2}(1+u^{2^k})$ by repeated difference of squares, and $1+u^{2^k}$ is $\Phi_{2^{k+1}}$. The factors of $1-u^8$ are the Φ_d with $d \mid 8$, and $3, 6 \nmid 8$. □

Theorem 3. *The rungs are the binary coordinates of the cube. $\prod_{k=0}^{2}(1+u^{2^k}) = \sum_{m=0}^{7} u^m$, each $m = 4b_2 + 2b_1 + b_0$ arising once, so $m \mapsto (b_2, b_1, b_0)$ is a bijection onto the corners $\{0,1\}^3$. On the poles $y = \sqrt{2}\,\omega$, with $\omega = e^{i\pi/4}$,*

$$\omega^m = (\omega^4)^{b_2}(\omega^2)^{b_1}\omega^{b_0}, \qquad \omega^4 = -1, \quad \omega^2 = i, \quad (5)$$

so the three bits are the half-turn, the quarter-turn and the eighth-turn, and flipping the most significant bit multiplies every pole character by -1.

Proof. Each $m < 8$ has one binary expansion; expand the product and the power. □

In (2) slot j carries the character ω^j, so Theorem 3 is the place-value reading of slot j as corner j mod 8 used in [4]. The ladder fixes which bit is which rung. It does not fix which spatial axis carries which bit, and the selection of BBP does not depend on that choice; whether the slot is counted as j or $j-1$ does matter [4].

Every Formula for π Drops the Coupling Row's Rung

Theorem 4. *If w is a formula of type $(*)$ for π, then $(y^2+2)(y^2+2y+2)$ divides p_w. Here $y^2+2 = 2\Phi_4(u)$, the Field Origin rung at the coupling row's level, and $y^2+2y+2 = 2(u^2+\sqrt{2}u+1)$, the half of $\Phi_8 = (u^2-\sqrt{2}u+1)(u^2+\sqrt{2}u+1)$ with poles at $135°$ and $225°$. The numerators are exactly*

$$p_{w(r)} = 16\,(y^2+2)(y^2+2y+2) \times \big[r\,(y^2-4y+2) - (y-1)\big]. \quad (6)$$

Proof. By (2), $\sum_j w_j S_j = -\frac{1}{8}\sum_\omega P_w(\sqrt{2}\omega)\,L_\omega$ with $P_w(z) = \sum_j w_j z^j = z\,p_w(z)/16$. Write $\omega_n = e^{i\pi n/4}$. Of the seven logarithms, $\log(\sqrt{2}+i)$ occurs only in L_{ω_2}, $\log(\sqrt{2}-i)$ only in L_{ω_6}, $\log(2-i)$ only in L_{ω_3} and $\log(2+i)$ only in L_{ω_5}, each with coefficient 1 [4]. The sum equals $\pi = -i \cdot i\pi$, which has no component on these four logarithms, and the coefficients $-\frac{1}{8}P_w(\sqrt{2}\omega_n)$ are algebraic. By the independence over the algebraic numbers, each is zero. So p_w vanishes at $\sqrt{2}\omega_n = \pm i\sqrt{2}$ and $-1 \pm i$, the roots of y^2+2 and y^2+2y+2. Dividing the numerators of $w_{\mathrm{BBP}} + r\,a$ gives (6). □

The reason is arithmetic. Over each conjugate pair of poles, the product of $1 - \bar{\omega}/\sqrt{2}$ is $\frac{1}{2}$ at $0°$ and $180°$, $\frac{1}{2}$ at $45°$ and $315°$, $\frac{3}{2}$ on the Φ_4 pair and $\frac{5}{2}$ on the $135°$ pair. Only the last two contain an odd prime, $3 = (\sqrt{2}+i)(\sqrt{2}-i)$ and $5 = (2+i)(2-i)$, and logarithms of those primes cannot combine into π. The cancellation is forced on the whole family; it is not particular to BBP.

π Is Read One Rung Up

Theorem 5. *After the cancellation in Theorem 4 the integrand of every formula for π is*

$$\frac{16\big[(y-1) - r\,(y^2-4y+2)\big]}{(y^2-2)(y^2-2y+2)}, \quad (7)$$

whose poles are $\pm\sqrt{2}$, on $\Phi_1\Phi_2$, and $1 \pm i$, on the $45°$ half of Φ_8. Only the poles $1 \pm i$ contribute to π. For BBP the $1 \pm i$ part integrates to $\pi + \ln 4$ and the $\pm\sqrt{2}$ part to $-\ln 4$. The rung that carries π is $\Phi_8 = \Phi_4(X^2)$, the Field Origin one level above the coupling row.

Proof. The integrand follows from (6). At $\pm\sqrt{2}$, $1 - \bar{\omega}/\sqrt{2} = 1 \mp 1/\sqrt{2}$ is real, so L_ω is real. At $1 \pm i$, $1 - \bar{\omega}/\sqrt{2} = (1 \pm i)/2$, whose logarithm is $-\frac{1}{2}\log 2 \pm i\pi/4$. The two integrals are elementary. □

Corollary. *The TIME slot w_7 is the coefficient of y^6 in (6), $16r$; the sum of the weights is $p_{w(r)}(1)/16 = -15r$. So silence on the TIME slot and closure, weights summing to zero, are each the condition $r = 0$, which leaves the factor $-(y-1)$ and is BBP. The factor $y-1$ is the closure condition.*

Part II: the Dictionary and the Ladder

Theorem 6 (the rung dictionary). *The values of formulas of type $(*)$ span the seven constants π, $\ln 2$, $\ln 3$, $\ln 5$, $\arctan\frac{1}{2}$, $\sqrt{2}\ln(1+\sqrt{2})$ and $\sqrt{2}\arctan(1/\sqrt{2})$; for each, the formulas are $w_c + r\,a$. Group the poles into*

the pairs $0/180°$ *(*$\pm\sqrt{2}$*),* $45/315°$ *(*$1 \pm i$*),* $90/270°$ *(the Field Origin,* $\pm i\sqrt{2}$*) and* $135/225°$ *(*$-1 \pm i$*). Then a formula for* c *must use a pair (no member of its family vanishes there) exactly in these cases:*

pair	constants whose formulas must use it
$0/180°$	$\sqrt{2}\ln(1+\sqrt{2})$
$45/315°$	π, $\arctan\frac{1}{2}$
$90/270°$	$\ln 3$, $\sqrt{2}\arctan(1/\sqrt{2})$
$135/225°$	$\ln 5$, $\arctan\frac{1}{2}$

Every family drops the $90/270°$ *and* $135/225°$ *pairs unless the table says otherwise, and each remaining pair is dropped by exactly one member.*

Proof. The seven formulas, for example $\ln 3 = 2S_2 + \frac{1}{2}S_6$ and $\arctan\frac{1}{2} = S_2 - \frac{1}{4}S_6$, are checked exactly by expanding (2) in the seven logarithms of [4]. Each entry is then the divisibility of the numerators p_{w_c+ra} by the pair's quadratic, for all r, for one r, or for none. □

Each pair is used by the constants built from its own number $1 - \bar{\omega}/\sqrt{2}$: the real units at $0/180°$; the argument $\pi/4$ at $45/315°$; the norm 3 and argument $\arctan(1/\sqrt{2})$ of $(\sqrt{2} \pm i)/\sqrt{2}$ on the Field Origin pair; the norm 5 and argument of $(1 \pm i)(2 \mp i)/2$ at $135/225°$. So π and $\ln 3$ are complementary: every formula for π drops the Field Origin pair, and every formula for $\ln 3$ uses it.

Theorem 7 (π **on one rung).** *The member* $r = -\frac{1}{4}$ *of the family for* π *is* $(2, 2, 1, 0, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{4}, 0)$*, the base* -4 *formula* $\pi = \sum_k (-\frac{1}{4})^k \left[\frac{2}{4k+1} + \frac{2}{4k+2} + \frac{1}{4k+3}\right]$*. It is the only member with no poles at* $\pm\sqrt{2}$*. Its integrand is* $4/(y^2 - 2y + 2)$*, so* $\pi = 4\arctan 1$ *is read from the* $45°$ *half of* Φ_8 *alone, and its natural denominator* $4 + y^4 = 4\,\Phi_8(u)$ *is a single rung.*

Proof. The bracket in (7) vanishes at $y = \pm\sqrt{2}$ exactly when $1 + 4r = 0$; the rest is substitution. □

BBP is the same π spread over the period-8 ladder: it adds the $\pm\sqrt{2}$ poles, which contribute $-\ln 4$ and cancel the logarithm from $1 \pm i$, and closure fixes the member.

Lemma 2. *Let* $k \geq 4$ *and let* ω *be a primitive* 2^k*-th root of unity. The conjugates of* $\alpha_\omega = 1 - \bar{\omega}/\sqrt{2} \in \mathbb{Q}(\zeta_{2^k})$ *are the* $\alpha_{\omega'}$ *for all primitive* ω'*, and*

$$N(\alpha_\omega) = \Phi_{2^k}\big(1/\sqrt{2}\big) = 1 + 2^{-2^{k-2}}. \qquad (8)$$

For $k = 3$ *the orbit is only half of the rung, which is why* Φ_8 *splits into its* $45°$ *and* $135°$ *halves.*

Proof. The automorphism $\zeta \mapsto \zeta^t$, t odd, sends $\sqrt{2}$ to $\varepsilon_t\sqrt{2}$ with $\varepsilon_t = \pm 1$ depending on t mod 8, and α_ω to $\alpha_{\varepsilon_t\omega^t}$. Since $-1 = \omega^{2^{k-1}}$, the index of $\varepsilon_t\omega^t$ is t or $t + 2^{k-1}$, which for $k \geq 4$ has the same residue mod 8 as t; so the indices run over all odd residues once. The product over all primitive ω' is $\Phi_{2^k}(1/\sqrt{2})$, because Φ_{2^k} is palindromic. □

Theorem 8 (the rungs carry the Field Origin column). *For* $k \geq 2$*,*

$$2^{2^{k-2}}\,\Phi_{2^k}\big(1/\sqrt{2}\big) = F_{k-2} = \Phi_4\big(K^{2^{k-4}}\big), \qquad (9)$$

where $F_j = 2^{2^j} + 1$ *is the* j*-th Fermat number and* $K = 4$*. So BBP's rungs* $\Phi_4, \Phi_8, \Phi_{16}, \Phi_{32}, \ldots$*, read at its own point* $1/\sqrt{2}$*, carry* $3, 5, 17, 257, \ldots$*: the Field Origin column at* $X = K^{1/4}, K^{1/2}, K, K^2, \ldots$*. The rung* Φ_{16} *carries the Field Origin's own value* $f(K, 0) = 17$*.*

Proof. $\Phi_{2^k}(x) = x^{2^{k-1}} + 1$ and $K^{2^{k-4}} = 2^{2^{k-3}}$. □

At level -1 the Field Origin value is $K + 1$ for every K. At level -2 it is $\sqrt{K} + 1$, which equals $D = K - 1$ only at $K = 4$; the primes 3 and 5 of Theorem 4 are these two values.

Theorem 9 (ladder invariance). *For* $m \geq 3$ *let type* $(*_m)$ *be base* $2^{2^{m-1}}$ *and period* 2^m*, whose poles again lie at radius* $\sqrt{2}$*. Suppose each* F_j*,* $2 \leq j \leq m - 2$*, has a prime factor that divides it exactly once; this holds for* $m \leq 6$*, since* F_2, F_3, F_4 *are prime. Then the numerator of every formula of type* $(*_m)$ *for* π *is divisible by* $y^2 + 2$*,* $y^2 + 2y + 2$ *and every rung* $y^{2^{k-1}} + 2^{2^{k-2}}$*,* $4 \leq k \leq m$*; its integrand is (7); and the formulas for* π *are the same one-parameter family at every level.*

Proof. Let p divide F_{k-2} exactly once. Every prime factor of F_{k-2} is $\equiv 1 \pmod{2^k}$ (Euler and Lucas [6]), so p splits completely in $\mathbb{Q}(\zeta_{2^k})$ into primes of norm p. By Lemma 2, $v_p(N(\alpha_\omega)) = 1$, so exactly one prime $\mathfrak{p}_\omega$ above p divides α_ω, and the $\mathfrak{p}_\omega$ of the rung are distinct conjugates. The Fermat numbers are pairwise coprime, so no other pole's α is divisible by a prime above p. Hence $\log\alpha_\omega$ is not in the $\mathbb{Q}$-span of $\log(-1)$ and the other $\log\alpha_{\omega'}$, and by Baker's theorem [3] not in their span over the algebraic numbers. A formula with value π therefore has zero coefficient on it: its numerator vanishes at $\sqrt{2}\omega$. The pairs at $90/270°$ and $135/225°$ keep their private primes above 3 and 5, as in Theorem 4. The forced factors leave a numerator of the form (forced) $\times\, Q(y)$ with $\deg Q \leq 3$, so the integrand is $Q(y)/\big((y^2 - 2)(y^2 - 2y + 2)\big)$, as at period 8,

and the same three conditions (no $\log 2$, no $\log(1+\sqrt{2})$, value π) select the same affine line. □

Climbing the ladder never adds a route to π. Each new rung carries a new Fermat prime, the first being the Field Origin's 17, and every formula for π must drop it. A blind search by lattice reduction at periods 16, 32 and 64, which does not use this theorem, finds exactly two integer relations among π and the slot series at each level, both in the lifted family; the next vectors of the reduced lattice have coefficients near 10^{22}.

Scope

The results are exact. The coupling row (Φ_6, Φ_4, Φ_3) at K and BBP's Field Origin column (Φ_2, Φ_4, Φ_8) meet only in Φ_4. Every formula of type $(*)$ for π drops that rung, and π is read from the rung above it.

The type $(*)$, base $16 = K^2$ and period $8 = 2^D$, is an input taken from [4]; the Lemma shows that it is the type whose poles reach $1 + i$. Nothing here says that BBP is the stellated octahedron. No symmetry of the stellated octahedron has order 8, so BBP's period is not one of its symmetries; the ladder places that period one rung up. The primes 3 and 5 are not given a meaning in Gravity Field Dynamics beyond Theorem 8, which identifies them as Field Origin values below the photon's level. Theorem 9 is proved for periods up to 64; beyond that it rests on each Fermat number having a prime factor to the first power, which is not known in general. The rungs at $X = K$ multiply to $5 \cdot 17 \cdot 257 = \sum_{m<8} K^m$, while $f(K, +1) = \sum_{m<3} K^m$; that is the geometric series, recorded only as an observation.

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